

THE GENERAL SUBGROUP PERMANENTAL-DOMINANCE CONJECTURE IN ORDER FOUR*

SIWEI ZENG[†]

Abstract. The general subgroup permanental-dominance conjecture was previously known only through matrix order three. This paper proves its complete order-four case: for every subgroup $H \leq \mathfrak{S}_4$, every irreducible complex character χ of H , and every 4×4 Hermitian positive-semidefinite matrix A , it establishes $d_\chi^H(A)/\chi(1) \leq \text{per } A$. Unlike the usual immanant specialization, the result covers all thirty-seven irreducible-character cases arising from the eleven conjugacy classes of subgroups of $\mathfrak{S}_4$. Thirty-five cases follow from general principal-minor, moment, and block-contraction inequalities. The two non-real A_4 characters are reduced to polynomial nonnegativity on the cone of 3×3 positive-semidefinite Gram matrices and are resolved by a rank-one sum-of-squares identity, an exact positive-definite interior certificate, rational Gram certificates, and closure of the Gram cone.

Key words. permanent, generalized matrix function, immanant, positive-semidefinite matrix, character theory, Gram matrix

AMS subject classifications. 15A15, 15B57, 20C15

1. Introduction and main theorem. The general subgroup permanental-dominance conjecture asks whether every normalized generalized matrix function associated with an irreducible character of every subgroup of $\mathfrak{S}_n$ is bounded above by the permanent on the Hermitian positive-semidefinite cone. Before the present work, this form of the conjecture was known only through matrix order three [17, 18]. The main result below establishes the complete order-four case. It is not merely the five-character $\mathfrak{S}_4$ immanant specialization: the eleven conjugacy classes of subgroups of $\mathfrak{S}_4$ produce thirty-seven irreducible-character cases, including two non-real A_4 characters that require a new Gram-cone argument.

The determinant and permanent are the two best-known members of the family of generalized matrix functions. Schur's classical theorem places the determinant below every normalized irreducible immanant on the Hermitian positive-semidefinite cone [15]. Marcus proved the permanent analogue of Hadamard's inequality [8], and Lieb strengthened that direction through a block inequality and conjectured that the permanent lies above every normalized immanant [7]. The formulation became known as the permanental-dominance conjecture through the work of Merris [9]. In its subgroup form, for an irreducible character χ of $H \leq \mathfrak{S}_n$, the conjectured inequality is

$$\frac{1}{\chi(1)} \sum_{\sigma \in H} \chi(\sigma) \prod_{i=1}^n A_{i, \sigma(i)} \leq \text{per } A \quad (A \succeq 0).$$

Taking $H = \mathfrak{S}_n$ recovers the usual irreducible-immanant statement. This all-subgroup conjecture is distinct from the stronger permanent-on-top conjecture, which asserts that the permanent is the largest eigenvalue of the Schur power matrix and is known to be false in general [18].

The conjecture has generated a substantial literature on dominance orderings and sufficient analytic

*Received by the editors on Month/Day/Year. Accepted for publication on Month/Day/Year. Handling Editor: Name of Handling Editor. Corresponding Author: Siwei Zeng.

[†]Independent Researcher, Beijing, China (endlesslethe@gmail.com). ORCID: 0009-0008-8914-7143.

mechanisms. Representative advances include identities and inequalities for generalized matrix functions [10], the ordering of hook immanants [4], row-appending and tensor methods [11, 12, 13], and the spectral approach of Soules [16]. Broader accounts and historical surveys are given by Cheon and Wanless [2], Zhang [19], and Wanless [18]. Rodtes obtained further identities and explicit sufficient criteria for classes of positive-semidefinite matrices [14]. More recent work relates immanant inequalities to weight-space methods [6], while the immanant form of permanental dominance has acquired an operational interpretation through generalized bosonic bunching probabilities [3].

Most of the literature concerns the immanant specialization $H = \mathfrak{S}_n$. That specialization is known in several broad families and, in particular, through order thirteen; these results do not by themselves cover characters of proper subgroups [12, 19, 18]. For matrix order $n = 3$, Tabata completed the description of the possible positions of normalized generalized matrix functions, including the remaining proper-subgroup cases [17]. In order four, Barrett, Hall, and Loewy characterized the cone of class-function inequalities and hence the inequalities among linear combinations of $\mathfrak{S}_4$ -immanants [1]. James also exhibited a positive-semidefinite matrix giving equality for one of the non-real A_4 characters [5, 2, 18].

These order-four results do not settle the full subgroup problem. A character of a proper subgroup, extended by zero to $\mathfrak{S}_4$, need not be a class function on $\mathfrak{S}_4$; in particular, odd conjugation interchanges the two non-real characters of A_4 . The distinction produces a substantially larger finite problem: $\mathfrak{S}_4$ has five irreducible characters, whereas all subgroups of $\mathfrak{S}_4$ contribute thirty-seven irreducible-character cases. The theorem below closes this order-four gap.

For $n \geq 1$, $H \leq \mathfrak{S}_n$, a character $\chi : H \rightarrow \mathbb{C}$, and $A = (A_{ij}) \in M_n(\mathbb{C})$, write

$$(1.1) \quad d_\chi^H(A) := \sum_{\sigma \in H} \chi(\sigma) \prod_{i=1}^n A_{i,\sigma(i)}, \quad \text{per } A := \sum_{\sigma \in \mathfrak{S}_n} \prod_{i=1}^n A_{i,\sigma(i)}.$$

The normalization in the conjecture is $d_\chi^H/\chi(1)$; no complex conjugate is placed on $\chi(\sigma)$ in (1.1).

THEOREM 1.1 (Permanental dominance in order four). *Let $H \leq \mathfrak{S}_4$, let V be an irreducible finite-dimensional complex representation of H , and let χ be its character. For every $A \in M_4(\mathbb{C})$ with $A \succeq 0$,*

$$(1.2) \quad \frac{d_\chi^H(A)}{\dim V} \leq \text{per } A.$$

Theorem 1.1 concerns matrix order four only; it makes no assertion in arbitrary order.

The proof has three deliberately separated layers. Its conceptual step is the Gram-cone reduction for the two non-real A_4 characters. Its completeness layer classifies all thirty-seven irreducible-character cases and resolves the other thirty-five by general inequalities. Its verification layer consists of exact polynomial certificates, supplemented by a Lean formalization. The latter two layers certify the scope and the algebra without obscuring the geometric mechanism.

The organizing idea is to write the exceptional gap as

$$c_0(u, v, w) + q^* K(u, v, w) q.$$

The block positivity constraint says that $B(u, v, w) - qq^*$ is a Gram matrix. After dividing out the nonzero coordinates of q , this residual becomes the Gram matrix R of three vectors, and the exceptional gap factors as

$$|q_1 q_2 q_3|^2 \mathcal{P}(R).$$

The problem is therefore moved from a character sum on four indices to the nonnegativity of one explicit polynomial on the cone of 3×3 Gram matrices.

Section 2 performs the analytic reductions and proves that the finite ledger is complete. Section 3 settles the thirty-five cases governed by general inequalities. Section 4 converts the exceptional character gap to the Gram polynomial, and Section 5 proves its nonnegativity and completes Theorem 1.1. Section 6 records an adjugate-matrix consequence. A contains the complete case data, B contains the exact coefficient certificates, and C maps the paper proof to its Lean counterparts.

2. Reductions and the finite case ledger.

2.1. Reality and reduction to correlation matrices.

LEMMA 2.1 (Reality). *If $A = A^*$ and χ is the character of a finite-dimensional complex representation of H , then $d_\chi^H(A)$ and $\text{per } A$ are real.*

Proof. Every finite-group representation may be made unitary, and hence $\overline{\chi(\sigma)} = \chi(\sigma^{-1})$. Hermitian symmetry gives

$$\overline{\prod_i A_{i,\sigma(i)}} = \prod_i A_{\sigma(i),i} = \prod_j A_{j,\sigma^{-1}(j)}.$$

Conjugating the sum and replacing σ by σ^{-1} proves the first claim. The second is the special case $H = \mathfrak{S}_n$, $\chi = 1$. $\square$

The normalization step is elementary but important because it removes all diagonal parameters without an invertibility hypothesis.

LEMMA 2.2 (Diagonal congruence). *Let $s_1, \dots, s_n \in \mathbb{R}$, and let*

$$D = \text{diag}(s_1, \dots, s_n).$$

Then

$$d_\chi^H(DAD) = \left(\prod_i s_i \right)^2 d_\chi^H(A), \quad \text{per}(DAD) = \left(\prod_i s_i \right)^2 \text{per } A.$$

Proof. For every permutation σ ,

$$\prod_i (DAD)_{i,\sigma(i)} = \prod_i (s_i A_{i,\sigma(i)} s_{\sigma(i)}) = \left(\prod_i s_i \right)^2 \prod_i A_{i,\sigma(i)}.$$

Summation proves both identities. $\square$

LEMMA 2.3 (Correlation reduction). *It is enough to prove Theorem 1.1 for positive-semidefinite matrices whose diagonal entries are all one.*

Proof. Suppose first that $A_{ii} > 0$ for all i and put $s_i = \sqrt{A_{ii}}$. Since a Hermitian matrix has real diagonal, the matrix

$$C = \text{diag}(s_i^{-1}) A \text{diag}(s_i^{-1})$$

is positive semidefinite and has diagonal one. Moreover,

$$A = \text{diag}(s_i) C \text{diag}(s_i).$$

Thus Lemma 2.2 transfers the desired inequality from C to A .

If $A_{ii} = 0$ for some i , positivity of the 2×2 principal minors implies $A_{ij} = A_{ji} = 0$ for every j . Each permutation monomial then contains a zero entry in row i , so both sides of (1.2) vanish. $\square$

We therefore write a general 4×4 correlation matrix as

$$(2.3) \quad A(a, b, c, d, e, f) = \begin{pmatrix} 1 & a & b & c \\ \bar{a} & 1 & d & e \\ \bar{b} & \bar{d} & 1 & f \\ \bar{c} & \bar{e} & \bar{f} & 1 \end{pmatrix} \succeq 0.$$

2.2. Subgroups and characters. Table 1 lists the eleven conjugacy types of subgroups of $\mathfrak{S}_4$. Here the superscripts t, d, n denote transposition, double transposition, and normal, respectively; in the dihedral row, $r = (1234)$ and $s = (13)$. The case column in Table 1 sums to

$$1 + 2 + 2 + 3 + 4 + 4 + 4 + 3 + 5 + 4 + 5 = 37.$$

Proposition 3.3 resolves thirty-five cases, and Proposition 5.6 resolves the two non-real A_4 cases.

LEMMA 2.4 (Subgroup classification). *Every subgroup of $\mathfrak{S}_4$ is conjugate to exactly one of the eleven types in Table 1.*

Proof. An intransitive subgroup preserves its orbit partition. For partition $2 + 1 + 1$ it is contained in S_2 and gives 1 or C_2^t . For $2 + 2$ it is contained in $S_2 \times S_2$ and gives, up to conjugacy, 1, C_2^t , C_2^d , or V_4^d . For $3 + 1$ it is contained in the displayed point stabilizer S_3 and gives 1, C_2^t , C_3 , or S_3 . This accounts for every intransitive type.

If the action is transitive, orbit-stabilizer shows that 4 divides $|H|$, while Lagrange's theorem gives $|H| \mid 24$. A transitive group of order four is regular and is either C_4 or the normal double-transposition group V_4^n . A group of order eight is a Sylow 2-subgroup, hence is conjugate to $\langle (1234), (13) \rangle \cong D_8$. The cases of order twelve and twenty-four are A_4 and S_4 , respectively. The listed types are distinguished by order, orbit structure, and, for the two order-two and two Klein-four types, by cycle type. $\square$

TABLE 1
Finite ledger of the irreducible-character cases for subgroups of $\mathfrak{S}_4$.

type	representative	group	$ H $	cases	resolution
1	$\{1\}$	1	1	1	principal permanent
C_2^t	$\langle (12) \rangle$	C_2	2	2	principal permanent
C_2^d	$\langle (12)(34) \rangle$	C_2	2	2	block contraction
C_3	$\langle (123) \rangle$	C_3	3	3	scalar triangle bounds
C_4	$\langle (1234) \rangle$	C_4	4	4	scalar and block bounds
V_4^n	$\{1, (12)(34), (13)(24), (14)(23)\}$	C_2^2	4	4	moment and scalar bounds
V_4^d	$\langle (12), (34) \rangle$	C_2^2	4	4	block contraction
S_3	$\text{Stab}(4)$	S_3	6	3	principal and scalar bounds
D_8	$\langle r, s \rangle$	D_8	8	5	block contractions
A_4	A_4	A_4	12	4	general bounds (2); Gram cone (2)
S_4	S_4	S_4	24	5	moment and scalar bounds

The character data used below are recorded in [A](#). Orthogonality of the character-table rows and the identity $\sum_{\chi} \chi(1)^2 = |H|$ prove that no irreducible character is missing. For the two exceptional A_4 characters we fix

$$(2.4) \quad \omega = -\frac{1}{2} - \frac{\sqrt{3}}{2}i, \quad \omega^2 + \omega + 1 = 0.$$

2.3. Conjugation and finite reduction. Simultaneous relabeling transports both the subgroup and its character data.

LEMMA 2.5 (Transport). *Let $g \in \mathfrak{S}_4$, $K = g^{-1}Hg$, and let $\chi^g(g^{-1}\sigma g) = \chi(\sigma)$. If $A_{ij}^g = A_{g(i),g(j)}$, then*

$$d_{\chi^g}^K(A^g) = d_{\chi}^H(A), \quad \text{per } A^g = \text{per } A, \quad A \succeq 0 \implies A^g \succeq 0.$$

Proof. Simultaneous permutation of rows and columns preserves positivity and the permanent. For a permutation monomial, changing i to $g(i)$ gives

$$\prod_i A_{i,(g^{-1}\sigma g)(i)}^g = \prod_j A_{j,\sigma(j)}.$$

Summing proves the assertion. $\square$

Thus it suffices to prove dominance in the thirty-seven irreducible-character cases associated with the eleven fixed representatives.

3. General inequalities and the thirty-five cases.

3.1. Four cycle aggregates. For (2.3) define

$$(3.5) \quad T = |a|^2 + |b|^2 + |c|^2 + |d|^2 + |e|^2 + |f|^2,$$

$$(3.6) \quad D = |a|^2|f|^2 + |b|^2|e|^2 + |c|^2|d|^2,$$

$$(3.7) \quad C = 2 \operatorname{Re}(adb + ae\bar{c} + bf\bar{c} + df\bar{e}),$$

$$(3.8) \quad F = 2 \operatorname{Re}(adf\bar{c} + ae\bar{f}b + b\bar{d}e\bar{c}).$$

Enumerating the cycle types of $\mathfrak{S}_4$ gives

$$(3.9) \quad \text{per } A = 1 + T + D + C + F.$$

The five normalized S_4 generalized matrix functions, in the order of the characters in (A.53), are

$$(3.10) \quad \begin{aligned} p_{[4]} &= 1 + T + D + C + F, \\ p_{[1^4]} &= 1 - T + D + C - F, \\ p_{[31]} &= 1 + \frac{1}{3}T - \frac{1}{3}D - \frac{1}{3}F, \\ p_{[211]} &= 1 - \frac{1}{3}T - \frac{1}{3}D + \frac{1}{3}F, \\ p_{[22]} &= 1 + D - \frac{1}{2}C. \end{aligned}$$

LEMMA 3.1 (Scalar correlation inequalities). *If A is the correlation matrix in (2.3), then*

$$(3.11) \quad 0 \leq T, \quad 0 \leq D, \quad D \leq \frac{T^2}{4}, \quad -2D \leq F \leq 2D \leq T,$$

$$(3.12) \quad \frac{T^2}{2} \leq T + \frac{3}{2}C, \quad 1 + D \leq \text{per } A.$$

In addition, if

$$(3.13) \quad U = |a|^2 + |b|^2 + |d|^2, \quad z = adb, \quad \bar{z} = adb,$$

then

$$(3.14) \quad U + 3 \operatorname{Re} z - \sqrt{3} \operatorname{Im} z \geq 0, \quad U + 3 \operatorname{Re} z + \sqrt{3} \operatorname{Im} z \geq 0.$$

Every 3×3 principal permanent is at most $\operatorname{per} A$.

Proof. The inequalities $|a|^2, \dots, |f|^2 \leq 1$ follow from the 2×2 principal minors. The first four estimates in (3.11) are then AM–GM and $2|\operatorname{Re}(xy)| \leq |x|^2 + |y|^2$, paired over opposite edges.

For the moment inequality, let $\lambda_1, \dots, \lambda_4 \geq 0$ be the eigenvalues of A . Direct trace expansion gives

$$\sum \lambda_i = 4, \quad \sum \lambda_i^2 = 4 + 2T, \quad \sum \lambda_i^3 = 4 + 6T + 3C.$$

The exact identity

$$\left(\sum_i \lambda_i \right) \left(\sum_i \lambda_i^3 \right) - \left(\sum_i \lambda_i^2 \right)^2 = \sum_{i < j} \lambda_i \lambda_j (\lambda_i - \lambda_j)^2$$

implies the first inequality in (3.12). It also gives $C \geq (T^2 - 2T)/3$. To prove the second inequality, subtract $1 + D$ from (3.9). If $T \leq 2$, then

$$T + C + F \geq T + \frac{T^2 - 2T}{3} - 2D \geq \frac{T(2 - T)}{6} \geq 0.$$

If $T \geq 2$, the moment bound gives $C \geq 0$, while $2D \leq T$ and $F \geq -2D$ give the same conclusion. In particular, $1 \leq \operatorname{per} A$.

We next prove (3.14). Put $u = |a|^2$, $v = |b|^2$, $w = |d|^2$, $S = u + v + w$, and write $z = x + iy$. Positivity of the principal 3×3 block gives

$$0 \leq u, v, w \leq 1, \quad x^2 + y^2 = uvw, \quad 1 - S + 2x \geq 0.$$

Also $27uvw \leq S^3$ by AM–GM. If $S \leq 9/4$, Cauchy–Schwarz gives

$$(3x \pm \sqrt{3}y)^2 \leq 12(x^2 + y^2) \leq \frac{4}{9}S^3 \leq S^2,$$

which proves both inequalities. If $S \geq 9/4$, then $x \geq (S - 1)/2$ and $|y| \leq 1$, so

$$S + 3x \pm \sqrt{3}y \geq \frac{5S - 3}{2} - \sqrt{3} > 0.$$

It remains to prove the principal-permanent contraction. Choose Gram vectors $x_1, \dots, x_4$ for A , write $x \odot y = x \otimes y + y \otimes x$, and set

$$y_1 = x_2 \odot x_3, \quad y_2 = x_1 \odot x_3, \quad y_3 = x_1 \odot x_2, \quad v = \bar{c}y_1 + \bar{e}y_2 + \bar{f}y_3.$$

The identity

$$\langle x \odot y, z \odot w \rangle = 2(\langle x, z \rangle \langle y, w \rangle + \langle x, w \rangle \langle y, z \rangle)$$

and direct collection of the cycle types give

$$\|v\|^2 = 2(\operatorname{per} A - \operatorname{per} A[\{1, 2, 3\}]).$$

Hence the displayed principal permanent is at most $\operatorname{per} A$; simultaneous relabeling proves the other three cases. $\square$

We also need two symmetric $2 + 2$ contractions. Put

$$(3.15) \quad R_+ := 1 + |b|^2 + |e|^2 + D + 2 \operatorname{Re}(adf\bar{c}),$$

$$(3.16) \quad R_\wedge := (1 - |b|^2)(1 - |e|^2) + |af - c\bar{d}|^2.$$

LEMMA 3.2 (Block contractions). *If A is the correlation matrix in (2.3), then*

$$(3.17) \quad R_+ \leq \operatorname{per} A, \quad R_\wedge \leq \operatorname{per} A, \quad 1 + |a|^2|f|^2 + |c|^2|d|^2 \leq \operatorname{per} A.$$

Proof. First note the exact identity

$$(3.18) \quad R_+ = (1 + |b|^2)(1 + |e|^2) + |af + c\bar{d}|^2.$$

Put $r = (1 - |b|^2)^{1/2}$ and

$$y_0 = a + \bar{d}b, \quad y_1 = \bar{d}r, \quad g_0 = c + fb, \quad g_1 = fr.$$

If $Y = |y_0|^2 + |y_1|^2$, $Z = |g_0|^2 + |g_1|^2$, and $h = \bar{y}_0g_0 + \bar{y}_1g_1$, expanding the definitions gives the certificate

$$(3.19) \quad \operatorname{per} A - R_+ = Y + Z + 2 \operatorname{Re}(\bar{e}h).$$

Cauchy–Schwarz and $|e| \leq 1$ imply $|\operatorname{Re}(\bar{e}h)|^2 \leq YZ$. Since $Y + Z \geq 2\sqrt{YZ} \geq 2|\operatorname{Re}(\bar{e}h)|$, the right-hand side of (3.19) is nonnegative. This proves the first contraction.

For the exterior contraction, take Gram vectors $x_1, \dots, x_4$ for A . Exterior Cauchy–Schwarz applied to $x_1 \wedge x_3$ and $x_2 \wedge x_4$ gives

$$|af - c\bar{d}|^2 \leq (1 - |b|^2)(1 - |e|^2),$$

where the inner product of the two exterior tensors is $af - c\bar{d}$.

We need two consequences of the first contraction. Relabeling its split twice and using (3.18) yields $(1 + |a|^2)(1 + |f|^2) \leq \operatorname{per} A$ and $(1 + |c|^2)(1 + |d|^2) \leq \operatorname{per} A$. Put

$$x = |a|^2|f|^2, \quad y = |c|^2|d|^2, \quad P = \operatorname{per} A.$$

If $x \leq y$, then $x \leq |c|^2$ and $1 + x + y \leq (1 + |c|^2)(1 + |d|^2) \leq P$; if $y \leq x$, use the other matching bound. This proves the third inequality in (3.17).

Finally let $p = |b|^2$, $q = |e|^2$, and $s = |af - c\bar{d}|$. The exterior bound gives $s^2 \leq (1 - p)(1 - q)$, hence $0 \leq s \leq 1$. The triangle inequality gives $s \leq \sqrt{x} + \sqrt{y}$. The two matching bounds and AM–GM imply

$$(1 + \sqrt{x})^2 \leq P, \quad (1 + \sqrt{y})^2 \leq P.$$

Consequently $(1 + s/2)^2 \leq P$. Moreover

$$(1 - p)(1 - q) + s^2 \leq (1 + s/2)^2,$$

because $p + q - pq \geq 0$ and $s - 3s^2/4 \geq 0$. Therefore $R_\wedge \leq P$, completing the proof. $\square$

3.2. Character cases covered by the general inequalities. The following proposition records the finite analytic part of the proof. A records the normalized functions used here. The only cases not covered by the present proposition are the two non-real A_4 characters.

PROPOSITION 3.3 (The thirty-five cases resolved by general inequalities). *For every representative subgroup in Lemma 2.4, all normalized generalized matrix functions other than those associated with χ_ω and $\bar{\chi}_\omega$ of A_4 are bounded above by per A .*

Proof. First consider the cyclic, Klein-four, and S_3 cases. In the C_3 and S_3 cases put $z = adb$. The expressions in Table 2 follow by listing the elements of the corresponding subgroup. We give the inequalities explicitly. Set

$$U = |a|^2 + |b|^2 + |d|^2, \quad z = x + iy, \quad P_3 = 1 + U + 2x.$$

Lemma 3.1 gives $P_3 \leq \text{per } A$ and $U + 3x \pm \sqrt{3}y \geq 0$. Thus

$$P_3 - (1 - x \pm \sqrt{3}y) = U + 3x \mp \sqrt{3}y \geq 0,$$

which verifies the C_3 cases; the value $1 + 2x$ is at most P_3 because $U \geq 0$. For S_3 , the trivial case is P_3 , the sign case differs from P_3 by $-2U$, and the standard case satisfies $P_3 - (1 - x) = U + 3x \geq 0$, obtained by averaging the two triangle bounds.

The trivial and C_2^t cases are bounded by P_3 (or by 1 after choosing the negative sign). For C_2^d , the positive value is at most $1 + |a|^2|f|^2 + |c|^2|d|^2 \leq \text{per } A$, while the negative value is at most one. For V_4^n , the all-positive value is $1 + D \leq \text{per } A$ and each other sign pattern is no larger. For V_4^d , all four values are no larger than $(1 + |a|^2)(1 + |f|^2) \leq \text{per } A$.

For C_4 , put

$$r = |b|^2|e|^2, \quad z = adf\bar{c}.$$

Its four normalized functions are listed in (A.55). Put $x = |a|^2|f|^2$ and $y = |c|^2|d|^2$. The estimates

$$2|\text{Re } z|, \quad 2|\text{Im } z| \leq x + y$$

show that the two values containing $+r$ are at most $1 + r + x + y = 1 + D \leq \text{per } A$. The other two are at most $1 - r + x + y \leq 1 + x + y \leq \text{per } A$ by (3.17).

For D_8 , write $p = |b|^2$, $q = |e|^2$. Its five normalized functions are listed in (A.56). The first and fourth are (3.17). By (3.18), the second differs from R_+ only by replacing $|af + c\bar{d}|^2$ with $-|af + c\bar{d}|^2$, and hence is no larger. The third and fifth are at most 1, hence at most per A .

For S_4 , the sign case follows from $T + F \geq T - 2D \geq 0$. For the two remaining nontrivial estimates, direct subtraction gives

$$3(\text{per } A - p_{[31]}) = 2T + 4D + 3C + 4F \geq T^2 - 4D \geq 0,$$

$$2(\text{per } A - p_{[22]}) = 2T + 3C + 2F \geq T^2 - 4D \geq 0.$$

Here we used (3.11)–(3.12). Finally, $p_{[211]} \leq 1 \leq \text{per } A$ because $F \leq T$ and $D \geq 0$.

Finally, the trivial A_4 case is the average of the trivial and sign S_4 cases, while its three-dimensional case is the average of the $[31]$ and $[211]$ cases. This proves the proposition. $\square$

4. The exceptional A_4 case: reduction to the Gram cone.

4.1. The exceptional character gap. It remains to treat the two conjugate non-real characters of A_4 . Their common geometric content is isolated below through their permanent-dominance gap.

Let

$$(4.20) \quad B(u, v, w) = \begin{pmatrix} 1 & u & v \\ \bar{u} & 1 & w \\ \bar{v} & \bar{w} & 1 \end{pmatrix}.$$

Define

$$(4.21) \quad c_0(u, v, w) = |u|^2 + |v|^2 + |w|^2 + 2 \operatorname{Re}((1 - \omega)uw\bar{v})$$

and the Hermitian matrix

$$(4.22) \quad K(u, v, w) = \begin{pmatrix} 1 & (1 - \omega^2)u + v\bar{w} & uw + (1 - \omega)v \\ (1 - \omega)\bar{u} + w\bar{v} & 1 & (1 - \omega^2)w + v\bar{u} \\ \bar{u}\bar{w} + (1 - \omega^2)\bar{v} & (1 - \omega)\bar{w} + u\bar{v} & 1 \end{pmatrix}.$$

LEMMA 4.1 (Character expansion). *For the correlation matrix (2.3),*

$$(4.23) \quad \operatorname{per} A - d_{\chi_\omega}^{A_4}(A) = c_0(a, b, d) + \begin{pmatrix} \bar{c} & \bar{e} & \bar{f} \end{pmatrix} K(a, b, d) \begin{pmatrix} c \\ e \\ f \end{pmatrix}.$$

For $\bar{\chi}_\omega$, the same formula holds after an odd simultaneous relabeling of the four indices.

Proof. Choose the labeling of C_+ so that $\chi_\omega(C_+) = \omega$. Listing the twelve elements gives the expansion

$$\begin{aligned} d_{\chi_\omega}^{A_4}(A) &= 1 + |a|^2|f|^2 + |b|^2|e|^2 + |c|^2|d|^2 \\ &\quad + \omega(e\bar{d}\bar{f} + a\bar{d}\bar{b} + b\bar{f}\bar{c} + c\bar{a}\bar{e}) \\ &\quad + \omega^2(df\bar{e} + ae\bar{c} + b\bar{a}\bar{d} + c\bar{b}\bar{f}). \end{aligned}$$

The four monomials on the last line are the conjugates of those on the preceding line, in the inverse order, so the displayed quantity is real. Subtracting it from (3.9), using $\omega^2 = -1 - \omega$, and collecting the constant, diagonal, and off-diagonal terms in (c, e, f) yields (4.23). An odd permutation interchanges the two three-cycle classes and therefore exchanges χ_ω with its conjugate. $\square$

The remaining task is consequently the following four-vector inequality.

THEOREM 4.2 (Four-vector inequality). *Suppose*

$$(4.24) \quad \mathcal{A} = \begin{pmatrix} B(u, v, w) & q \\ q^* & 1 \end{pmatrix} \succeq 0, \quad q \in \mathbb{C}^3.$$

Then

$$(4.25) \quad c_0(u, v, w) + q^* K(u, v, w) q \geq 0.$$

The proof is developed in the residual normalization below and in Section 5.

4.2. Residual-vector normalization. There is a useful geometric meaning behind the next change of variables. Realize the block matrix (4.24) as the Gram matrix of four unit vectors. The vector corresponding to the last row and column determines a common one-dimensional direction. Subtracting the projections of the first three vectors onto that direction leaves three residual vectors, whose Gram matrix is $B - qq^*$. Dividing the residual vectors by the coordinates of the removed projections does not change positivity; it merely produces the scale-free matrix R below. This is the step that turns a four-vector completion problem into a problem on the cone of three-vector Gram matrices.

Assume first that $q_1 q_2 q_3 \neq 0$. The Schur complement of the final diagonal entry in (4.24) is $B - qq^* \succeq 0$. Congruence by $\text{diag}(\bar{q}_1^{-1}, \bar{q}_2^{-1}, \bar{q}_3^{-1})$ gives the residual Gram matrix

$$(4.26) \quad R = \begin{pmatrix} n_1 & h_{12} & h_{13} \\ \bar{h}_{12} & n_2 & h_{23} \\ \bar{h}_{13} & \bar{h}_{23} & n_3 \end{pmatrix} \succeq 0,$$

where

$$(4.27) \quad n_i = \frac{1 - |q_i|^2}{|q_i|^2}, \quad h_{12} = \frac{u}{q_1 \bar{q}_2} - 1, \quad h_{13} = \frac{v}{q_1 \bar{q}_3} - 1, \quad h_{23} = \frac{w}{q_2 \bar{q}_3} - 1.$$

Thus R is the Gram matrix of three residual vectors w_1, w_2, w_3 with $n_i = \|w_i\|^2$ and $h_{ij} = \langle w_i, w_j \rangle$.

We now define the normalized geometric polynomial. Put

$$(4.28) \quad \begin{aligned} E &= n_1 n_2 + |h_{12}|^2 + n_1 n_3 + |h_{13}|^2 + n_2 n_3 + |h_{23}|^2, \\ x_{12,13} &= n_1 h_{23} + h_{13} \bar{h}_{12}, \\ x_{12,23} &= h_{12} h_{23} + n_2 h_{13}, \\ x_{13,23} &= n_3 h_{12} + h_{13} \bar{h}_{23}, \\ S &= E + 2 \operatorname{Re}(x_{12,13} + x_{12,23} + x_{13,23}), \\ S_\omega &= E + 2 \operatorname{Re}(\omega^2 x_{12,13} + \omega x_{12,23} + \omega^2 x_{13,23}), \\ \tau &= h_{13} \bar{h}_{12} \bar{h}_{23}, \\ \Gamma &= n_3 |h_{12}|^2 + n_2 |h_{13}|^2 + n_1 |h_{23}|^2 + 3 \operatorname{Re} \tau + \sqrt{3} \operatorname{Im} \tau. \end{aligned}$$

Finally set

$$(4.29) \quad \mathcal{P}(R) = 24 + 6(n_1 + n_2 + n_3 + 2 \operatorname{Re}(h_{12} + h_{13} + h_{23})) + 2S - S_\omega + \Gamma.$$

LEMMA 4.3 (Exact normalization identity). *With (4.27),*

$$(4.30) \quad c_0(u, v, w) + q^* K(u, v, w) q = |q_1 q_2 q_3|^2 \mathcal{P}(R).$$

Proof. Equations (4.27) give

$$u = q_1 \bar{q}_2 (1 + h_{12}), \quad v = q_1 \bar{q}_3 (1 + h_{13}), \quad w = q_2 \bar{q}_3 (1 + h_{23}),$$

and $|q_i|^2(1 + n_i) = 1$. Let $Q = |q_1 q_2 q_3|^2$. Dividing first by Q organizes the cancellations by degree. The two exact quotient expansions are recorded in (B.58)–(B.59). They are obtained term by term; for example,

$$\frac{|u|^2}{Q} = (1 + n_3)|1 + h_{12}|^2, \quad \frac{u w \bar{v}}{Q} = (1 + h_{12})(1 + h_{23})(1 + \bar{h}_{13}).$$

Expanding the two quotient formulas and using $1 + \omega + \omega^2 = 0$ groups the degree-two terms as $2S - S_\omega$ and the degree-three terms as Γ ; the remaining terms are the constant and linear part in (4.29). Thus their sum is $\mathcal{P}(R)$, which proves (4.30). $\square$

Thus Theorem 4.2 follows from the next result.

THEOREM 4.4 (Geometric polynomial). *For every Hermitian matrix R of the form (4.26),*

$$R \succeq 0 \implies \mathcal{P}(R) \geq 0.$$

5. Positivity on the Gram cone. The proof uses the geometry of the Gram cone in three stages. On the rank-one locus the three residual vectors are collinear and the polynomial has a direct sum-of-squares form. In the positive-definite region, two residual vectors are used as coordinates and the third is minimized through an explicit two-dimensional Hermitian quadratic form. Finally, every singular Gram matrix is obtained as a limit of positive-definite ones. The rank-one calculation is also the algebraic base coefficient in the interior minimum certificate, so it is both explanatory and logically active.

5.1. The rank-one boundary. Suppose first that R has rank one. Write

$$n_1 = |x|^2, \quad n_2 = |y|^2, \quad n_3 = |z|^2, \quad h_{12} = \bar{x}y, \quad h_{13} = \bar{x}z, \quad h_{23} = \bar{y}z.$$

Define

$$(5.31) \quad R_0(x, y, z) = 4 + |x + y + z|^2 + |xy + xz + yz|^2 - (|xy|^2 + |xz|^2 + |yz|^2) + |xyz|^2,$$

$$(5.32) \quad \Phi(x, y, z) = xy + \omega xz + \omega^2 yz.$$

LEMMA 5.1 (Rank-one sum of squares). *For all $x, y, z \in \mathbb{C}$,*

$$(5.33) \quad \mathcal{P}(R) = 2(3R_0(x, y, z) + |\Phi(x, y, z)|^2) \geq 0.$$

Proof. Regard R_0 as a quadratic in z . Put

$$K_0 = |1 + \bar{x}y|^2, \quad L_0 = \overline{x + y} + \bar{x}\bar{y}(x + y), \quad C_0 = 4 + |x + y|^2.$$

Direct expansion gives

$$R_0 = K_0|z|^2 + 2\operatorname{Re}(L_0z) + C_0$$

and the exact discriminant identity

$$C_0K_0 - |L_0|^2 = 4(1 + \operatorname{Re}(\bar{x}y))^2 \geq 0.$$

If $K_0 > 0$, completing the square proves $R_0 \geq 0$. If $K_0 = 0$, the discriminant identity forces $L_0 = 0$, and then $R_0 = C_0 > 0$. A second direct expansion using $1 + \omega + \omega^2 = 0$ gives the equality in (5.33); its right-hand side is nonnegative. $\square$

5.2. The positive-definite region. It remains to treat a positive-definite residual Gram matrix. Write

$$(5.34) \quad a = n_1 > 0, \quad b = n_2, \quad c = n_3, \quad h_{12} = g = r - is, \quad p = h_{13}, \quad q = h_{23},$$

and put

$$(5.35) \quad G = \begin{pmatrix} a & g \\ \bar{g} & b \end{pmatrix}, \quad \delta = \det G = ab - r^2 - s^2 > 0, \quad h = \begin{pmatrix} p \\ q \end{pmatrix}.$$

There is a unique $\zeta \in \mathbb{C}^2$ such that $h = G\zeta$. Positivity of R tested on $(-\zeta, 1)$ gives the Schur slack

$$(5.36) \quad \zeta^* G \zeta \leq c.$$

Set

$$(5.37) \quad \lambda = 6 + a + b + r^2 + s^2 + 5r - \sqrt{3}s,$$

$$(5.38) \quad P_0 = 24 + 6(a + b + 2r) + ab + r^2 + s^2,$$

$$(5.39) \quad \ell = \begin{pmatrix} 6 + (2 - \omega)g + (2 - \omega^2)b \\ 6 + (2 - \omega)a + (2 - \omega^2)\bar{g} \end{pmatrix},$$

$$(5.40) \quad M = \begin{pmatrix} 1 + b & 2 - \omega + (1 - \omega)g \\ 2 - \omega^2 + (1 - \omega^2)\bar{g} & 1 + a \end{pmatrix},$$

$$(5.41) \quad \Theta = \lambda I_2 + MG.$$

LEMMA 5.2 (Quadratic decomposition). *With the preceding notation,*

$$(5.42) \quad \begin{aligned} \mathcal{P}(R) &= P_0 + 2 \operatorname{Re}(\ell^* h) + \lambda c + \operatorname{Re}(h^* M h) \\ &= Q(\zeta) + \lambda(c - \zeta^* G \zeta), \end{aligned}$$

where

$$(5.43) \quad Q(\zeta) = P_0 + 2 \operatorname{Re}((G\ell)^* \zeta) + \zeta^* (G\Theta) \zeta.$$

Moreover $G\Theta$ is Hermitian.

Proof. Substitution of (5.34) in (4.29) gives the first identity. Replacing h by $G\zeta$, using $G = G^*$ and $\Theta = \lambda I + MG$, gives the second. Finally $G\Theta = \lambda G + GMG$ is Hermitian because G and M are Hermitian. $\square$

LEMMA 5.3 (Positivity of the scalar part). *Under $a, b \geq 0$ and $r^2 + s^2 \leq ab$, one has $\lambda > 0$.*

Proof. Let $\rho = \sqrt{r^2 + s^2}$. Cauchy–Schwarz and AM–GM give

$$5r - \sqrt{3}s \geq -2\sqrt{7}\rho, \quad a + b \geq 2\rho.$$

Consequently

$$\lambda \geq (\rho - (\sqrt{7} - 1))^2 + 2(\sqrt{7} - 1) > 0. \quad \square$$

The remaining algebra is organized by the Gram defect δ . Define

$$(5.44) \quad N = \operatorname{Re}(P_0 \det \Theta - \ell^* G \operatorname{adj}(\Theta) \ell).$$

After the substitution $b = (r^2 + s^2 + \delta)/a$, multiplication by a^3 clears the denominators in $\operatorname{Re} \det \Theta$ and N . Their coefficient expansions are therefore polynomial identities in $\mathbb{Q}(\sqrt{3})[a, r, s, \delta]$. The representation of N_0 below temporarily introduces square roots only to expose its sign.

LEMMA 5.4 (Quadratic-form and minimum certificates). *For $a > 0$ and $\delta \geq 0$,*

$$(5.45) \quad a^3 \operatorname{Re} \det \Theta = D_0 + \delta D_1 + \delta^2 D_2,$$

where $D_0, D_1, D_2 > 0$. More precisely, $D_0 = a^3 \lambda_0 H_0$, where λ_0 denotes (5.37) with $b = (r^2 + s^2)/a$ and $H_0 = 2\lambda_0 - 6 + 4(r^2 + s^2) > 0$. Furthermore

$$(5.46) \quad a^3 N = N_0 + \delta N_1 + \delta^2 N_2 + \delta^3 N_3,$$

with $N_0 \geq 0$ and $N_1, N_2, N_3 > 0$.

Proof. Substitution of $b = (r^2 + s^2 + \delta)/a$ in (5.41) and (5.44) gives the two polynomial expansions. Their exact coefficients are recorded in B.2; here we isolate the geometric content of their signs.

Put $\rho = (r^2 + s^2)^{1/2}$ and $k = \sqrt{7} - 1$. On the boundary $\delta = 0$, the two estimates in Lemma 5.3 give

$$\lambda_0 \geq (\rho - k)^2 + 2k.$$

Consequently

$$(5.47) \quad H_0 \geq 6\rho^2 - 4k\rho + 6 = 6 \left(\rho - \frac{k}{3} \right)^2 + \frac{2 + 4\sqrt{7}}{3} > 0.$$

This proves $D_0 > 0$. The positive residual certificates for D_1 and D_2 are given in B.2.

It remains conceptually important to identify the constant coefficient in (5.46). Set $\delta = 0$. The first two residual vectors are then collinear. Define

$$x_0 = \sqrt{a}, \quad y_0 = \frac{g}{\sqrt{a}}, \quad L'_0 = x_0 \ell_1 + y_0 \ell_2,$$

where ℓ_1, ℓ_2 are the components of (5.39), evaluated at $b = (r^2 + s^2)/a$. Then $|x_0|^2 = a$, $|y_0|^2 = b$, and $\bar{x}_0 y_0 = g$. For $z \in \mathbb{C}$, let

$$R_{\text{rk1}}(z) = \begin{pmatrix} a & g & \bar{x}_0 z \\ \bar{g} & b & \bar{y}_0 z \\ x_0 \bar{z} & y_0 \bar{z} & |z|^2 \end{pmatrix}.$$

This is the rank-one Gram matrix generated by (x_0, y_0, z) . Substitution in (5.42) gives the exact scalar quadratic

$$(5.48) \quad \mathcal{P}(R_{\text{rk1}}(z)) = P_0 + 2 \operatorname{Re}(\overline{L'_0} z) + H_0 |z|^2.$$

Indeed, the linear term is $\ell^* \begin{pmatrix} \bar{x}_0 z & \bar{y}_0 z \end{pmatrix}^T = \overline{L'_0} z$, and the coefficient of $|z|^2$ is $H_0 = 2\lambda_0 - 6 + 4(r^2 + s^2)$. By Lemma 5.1, the left-hand side of (5.48) is nonnegative for every $z \in \mathbb{C}$. Since $H_0 > 0$, minimizing the right-hand side gives $|L'_0|^2 \leq P_0 H_0$. Substitution in (5.44) now yields the exact boundary coefficient

$$(5.49) \quad N_0 = a^3 \lambda_0 (P_0 H_0 - |L'_0|^2) \geq 0.$$

The exact human-readable certificates for N_1, N_2, N_3 occupy the remaining subsections of B. They show respectively that each remaining coefficient is strictly positive, without numerical approximation. $\square$

PROPOSITION 5.5 (Positive-definite residual Gram matrices). *If $R \succ 0$, then $\mathcal{P}(R) > 0$.*

Proof. Equations (5.45) and (B.60) show $\operatorname{Re} \det \Theta > 0$. Direct multiplication gives the exact leading-entry identity

$$(5.50) \quad (G\Theta)_{11} = aH_0 + \delta(a+1) > 0.$$

Moreover

$$\det(G\Theta) = \det G \det \Theta = \delta \det \Theta.$$

Because $G\Theta$ is Hermitian, its determinant is real; since $\delta > 0$, this also proves $\det \Theta = \operatorname{Re} \det \Theta$. Hence $\det(G\Theta) = \delta \operatorname{Re} \det \Theta > 0$. The two leading principal minors are positive, so the 2×2 Hermitian matrix $G\Theta$ is positive definite.

The critical-point equation for (5.43) is $G\Theta\zeta = -G\ell$. Since G is invertible, it is equivalent to $\Theta\zeta = -\ell$, and therefore

$$\zeta_0 = -\frac{\operatorname{adj}(\Theta)\ell}{\operatorname{Re} \det \Theta},$$

and completing the Hermitian square gives

$$\min_{\zeta} Q(\zeta) = \frac{N}{\operatorname{Re} \det \Theta}.$$

By (5.46), $N > 0$ when $\delta > 0$. Therefore $Q(\zeta) > 0$ for all ζ . Lemma 5.3 and the Schur slack (5.36) now give

$$\mathcal{P}(R) = Q(\zeta) + \lambda(c - \zeta^* G \zeta) > 0. \quad \square$$

5.3. Closure of the Gram cone.

Proof of Theorem 4.4. For $R \succeq 0$ and $t > 0$, the matrix $R+tI$ is positive definite, and hence $\mathcal{P}(R+tI) > 0$ by the preceding proposition. Since $\mathcal{P}$ is a polynomial in the real and imaginary parts of the entries, letting $t \downarrow 0$ gives $\mathcal{P}(R) \geq 0$. $\square$

Proof of Theorem 4.2. If all coordinates of q are nonzero, the residual matrix (4.26) is positive semidefinite. Theorem 4.4 and (4.30) give (4.25).

For arbitrary q , choose phases p_i of modulus one, taking $p_i = q_i/|q_i|$ when $q_i \neq 0$ and $p_i = 1$ otherwise. Convexly interpolate the block matrix (4.24) with the rank-one correlation completion generated by $(p_1, p_2, p_3, 1)$. Its new last column is $q_i(t) = (1-t)q_i + tp_i$, which is nonzero for $0 < t < 1$. The interpolated matrix remains positive semidefinite, so the already proved case applies. Letting $t \downarrow 0$ and using continuity proves the result. $\square$

5.4. Completion of the main theorem. We now combine the geometric inequality with the finite character classification.

PROPOSITION 5.6 (The two non-real A_4 characters). *For every 4×4 positive-semidefinite correlation matrix A ,*

$$d_{\chi_\omega}^{A_4}(A) \leq \operatorname{per} A, \quad d_{\bar{\chi}_\omega}^{A_4}(A) \leq \operatorname{per} A.$$

Proof. Write A in the block form (4.24) with $B = B(a, b, d)$ and $q = (c, e, f)^T$. Lemma 4.1 and Theorem 4.2 give the first inequality. An odd simultaneous relabeling preserves positivity and the permanent and interchanges the two non-real characters, giving the second. $\square$

Proof of Theorem 1.1. By Lemma 2.3, it is enough to consider correlation matrices. Lemma 2.4 and the transport Lemma 2.5 reduce an arbitrary subgroup and irreducible representation to one of the listed character cases. Proposition 3.3 proves thirty-five of them and Proposition 5.6 proves the remaining two. Lemma 2.1 shows that the compared quantities are real. Finally, Lemma 2.2 returns from correlation matrices to the full positive-semidefinite cone. $\square$

6. An adjugate-matrix consequence. The four-vector inequality has a useful matrix consequence. It is recorded after the proof of Theorem 1.1 to make clear that it is a consequence of the residual-vector argument, not an additional step in it.

LEMMA 6.1 (Ellipsoid-to-adjugate certificate). *Let $B, K \in M_m(\mathbb{C})$ with $B \succeq 0$ and $K = K^*$, and let $c \in \mathbb{R}$. If*

$$\begin{pmatrix} B & q \\ q^* & 1 \end{pmatrix} \succeq 0 \implies c + q^* K q \geq 0$$

for every $q \in \mathbb{C}^m$, then

$$(6.51) \quad (\det B)K + c \operatorname{adj}(B) \succeq 0.$$

Proof. If $\det B = 0$, the choice $q = 0$ gives $c \geq 0$ and $\operatorname{adj}(B) \succeq 0$, so (6.51) follows. To justify the latter fact, if $\operatorname{rank} B \leq m - 2$, then $\operatorname{adj}(B) = 0$; if $\operatorname{rank} B = m - 1$, spectral decomposition shows that $\operatorname{adj}(B)$ is the product of the positive eigenvalues times the orthogonal projection onto $\ker B$. If $\det B > 0$, then $B \succ 0$ and the block condition is $q^* B^{-1} q \leq 1$. Given $y \neq 0$, set

$$q = \frac{y}{\sqrt{y^* B^{-1} y}}.$$

The hypothesis, multiplied by $y^* B^{-1} y$, says $y^*(K + cB^{-1})y \geq 0$. Hence $K + cB^{-1} \succeq 0$. Multiplication by $\det B > 0$ and the identity $\operatorname{adj}(B) = (\det B)B^{-1}$ prove the claim. $\square$

Applied to (4.21)–(4.22), Lemma 6.1 gives the positive-semidefinite certificate

$$(6.52) \quad (\det B(u, v, w))K(u, v, w) + c_0(u, v, w) \operatorname{adj}(B(u, v, w)) \succeq 0.$$

Appendix A. Complete case data.

A.1. Character tables. This appendix records the non-cyclic character data used in the finite character reduction. For S_4 , use the conjugacy-class ordering

$$1, \quad (12), \quad (12)(34), \quad (123), \quad (1234).$$

Its irreducible character table is

$$(A.53) \quad \begin{array}{c|ccccc} & 1 & (12) & (12)(34) & (123) & (1234) \\ \hline [4] & 1 & 1 & 1 & 1 & 1 \\ [1^4] & 1 & -1 & 1 & 1 & -1 \\ [31] & 3 & 1 & -1 & 0 & -1 \\ [211] & 3 & -1 & -1 & 0 & 1 \\ [22] & 2 & 0 & 2 & -1 & 0 \end{array}$$

For A_4 , let C_+ and C_- be the two conjugacy classes of three-cycles. With ω as in (2.4), the table is

$$(A.54) \quad \begin{array}{c|cccc} & 1 & (12)(34) & C_+ & C_- \\ \hline 1 & 1 & 1 & 1 & 1 \\ \chi_\omega & 1 & 1 & \omega & \omega^2 \\ \bar{\chi}_\omega & 1 & 1 & \omega^2 & \omega \\ \chi_3 & 3 & -1 & 0 & 0 \end{array}$$

For $D_8 = \langle r, s : r^4 = s^2 = 1, srs = r^{-1} \rangle$, the four linear characters take independent signs on r and s , while its two-dimensional character is $2, -2, 0, 0, 0$ on

$$1, \quad r^2, \quad \{r, r^3\}, \quad \{s, r^2s\}, \quad \{rs, r^3s\}.$$

The standard character tables of C_m , C_2^2 , and S_3 supply the remaining cases listed in Section 3.

A.2. Normalized functions and case accounting. For the smaller subgroups, listing the permutation monomials gives Table 2. In the C_3 and S_3 rows, $z = adb$. In the V_4^n row, $r = |a|^2|f|^2$, $s = |b|^2|e|^2$, and $t = |c|^2|d|^2$; the four signs are the four character sign patterns of C_2^2 .

TABLE 2
Normalized generalized matrix functions for the smaller subgroup cases.

group	normalized generalized matrix functions
1	1
C_2^t	$1 \pm a ^2$
C_2^d	$1 \pm a ^2 f ^2$
C_3	$1 + 2 \operatorname{Re} z, \quad 1 - \operatorname{Re} z \pm \sqrt{3} \operatorname{Im} z$
V_4^n	$1 \pm r \pm s \pm t$ with the four character sign patterns
V_4^d	$(1 \pm a ^2)(1 \pm f ^2)$
S_3	$1 + a ^2 + b ^2 + d ^2 + 2 \operatorname{Re} z,$ $1 - a ^2 - b ^2 - d ^2 + 2 \operatorname{Re} z, \quad 1 - \operatorname{Re} z$

For C_4 , set $r = |b|^2|e|^2$ and $z = adf\bar{c}$. The four normalized functions are

$$(A.55) \quad 1 + r + 2 \operatorname{Re} z, \quad 1 - r - 2 \operatorname{Im} z, \quad 1 + r - 2 \operatorname{Re} z, \quad 1 - r + 2 \operatorname{Im} z.$$

For D_8 , set $p = |b|^2$ and $q = |e|^2$. Its five normalized functions are

$$(A.56) \quad \begin{aligned} &R_+, \quad (1+p)(1+q) - |af + c\bar{d}|^2, \\ &(1-p)(1-q) - |af - c\bar{d}|^2, \quad R_\wedge, \quad 1 - pq. \end{aligned}$$

The five S_4 functions are displayed in (3.10). For A_4 , the trivial and three-dimensional normalized functions are, respectively,

$$(A.57) \quad 1 + D + C, \quad 1 - \frac{1}{3}D,$$

while the two non-real functions are the conjugate pair represented by the expansion in Lemma 4.1. Thus the numbers of entries are

$$1 + 2 + 2 + 3 + 4 + 4 + 4 + 3 + 5 + 4 + 5 = 37,$$

with thirty-five covered by Proposition 3.3 and the remaining two by Proposition 5.6.

Appendix B. Exact algebraic certificates.

This appendix records the longer exact expansions and the non-obvious polynomial certificates used in the geometric proof. They contain no numerical approximation.

B.1. Residual-normalization expansion. With the notation of Section 4.2 and $Q = |q_1 q_2 q_3|^2$, direct substitution in (4.21) gives

$$(B.58) \quad \frac{c_0(u, v, w)}{Q} = (1 + n_3)|1 + h_{12}|^2 + (1 + n_2)|1 + h_{13}|^2 + (1 + n_1)|1 + h_{23}|^2 \\ + 2 \operatorname{Re}((1 - \omega)(1 + h_{12})(1 + h_{23})(1 + \bar{h}_{13})).$$

For the quadratic part, the diagonal terms and the three upper-triangular entries of K give

$$(B.59) \quad \frac{q^* K q}{Q} = (1 + n_2)(1 + n_3) + (1 + n_1)(1 + n_3) + (1 + n_1)(1 + n_2) \\ + 2 \operatorname{Re}\left((1 - \omega^2)(1 + n_3)(1 + h_{12}) + (1 + h_{13})(1 + \bar{h}_{23}) \right. \\ \left. + (1 + h_{12})(1 + h_{23}) + (1 - \omega)(1 + n_2)(1 + h_{13}) \right. \\ \left. + (1 - \omega^2)(1 + n_1)(1 + h_{23}) + (1 + h_{13})(1 + \bar{h}_{12})\right).$$

Collecting these two formulas by degree gives (4.30).

B.2. Determinant and numerator coefficients. After $b = (r^2 + s^2 + \delta)/a$, direct coefficient collection in (5.41) gives (5.45) with

$$(B.60) \quad D_0 = a^3 \lambda_0 H_0, \\ D_1 = a((4 + 11a)(r^2 + s^2) + a(20 + a)r \\ - \sqrt{3}a(4 + 3a)s + a(18 + 10a + 3a^2)),$$

$$(B.61) \quad D_2 = a(a + 1)(a + 2).$$

The sign of D_0 was proved in the main text. For D_1 , set

$$Q = 4 + 11a, \quad L_r = a(20 + a), \quad L_s = -\sqrt{3}a(4 + 3a), \quad C = a(18 + 10a + 3a^2).$$

The exact identity

$$(B.62) \quad 4Q(Q(r^2 + s^2) + L_r r + L_s s + C) = (2Qr + L_r)^2 + (2Qs + L_s)^2 \\ + 4QC - L_r^2 - L_s^2$$

reduces positivity to

$$4QC - L_r^2 - L_s^2 = 4a(72 + 126a + 94a^2 + 26a^3) > 0.$$

Thus $D_1 > 0$, and $D_2 > 0$ is immediate.

Coefficient collection in (5.44) gives (5.46). The constant coefficient is (5.49). At the other end,

$$N_3 = (a + 2)(a^2 + 6) > 0.$$

For N_2 , one obtains

$$\begin{aligned} N_2 &= 36(r^2 + s^2) + a(120 + 84r - 36\sqrt{3}s + 36(r^2 + s^2)) \\ &\quad + a^2(60 + 24r - 24\sqrt{3}s + 16(r^2 + s^2)) \\ &\quad + a^3(16 + 2(r^2 + s^2)) + 2a^4. \end{aligned}$$

Thus $N_2 = Q_2(r^2 + s^2) + L_{2r}r + L_{2s}s + C_2$, where

$$\begin{aligned} Q_2 &= 36 + 36a + 16a^2 + 2a^3, \\ L_{2r} &= 84a + 24a^2, \quad L_{2s} = -\sqrt{3}(36a + 24a^2), \\ C_2 &= 120a + 60a^2 + 16a^3 + 2a^4. \end{aligned}$$

Applying (B.62) gives

$$4Q_2C_2 - L_{2r}^2 - L_{2s}^2 = 16a(a^6 + 16a^5 + 112a^4 + 318a^3 + 588a^2 + 936a + 1080) > 0,$$

and hence $N_2 > 0$.

For N_1 , put $X = r$, $Y = \sqrt{3}s$ and $\nu = X^2 + Y^2/3$. Then

$$(B.63) \quad N_1 = 36\nu^2 + aB_1 + a^2B_2 + a^3B_3 + a^4B_4 + 6a^5,$$

where

$$\begin{aligned} B_1 &= \nu(240 + 168X - 72Y + 54\nu), \\ B_2 &= 432 + 624X - 240Y + 492X^2 + 108Y^2 - 168XY \\ &\quad + (156X - 84Y)\nu + 38\nu^2, \\ B_3 &= 168 + 72X - 168Y + 54X^2 + 46Y^2 - 84XY \\ &\quad + (20X - 24Y)\nu + 6\nu^2, \\ B_4 &= 60 + 24X - 24Y + 14\nu. \end{aligned}$$

The next two subsections prove the required signs.

B.3. Direct square completions. The coefficient B_2 has the exact decomposition

$$\begin{aligned} B_2 &= 38 \left(\nu + \frac{39}{19}X - \frac{21}{19}Y + \frac{3}{2} \right)^2 + \frac{448}{19} \left(Y + \frac{3}{32}X - \frac{1083}{448} \right)^2 \\ &\quad + \frac{3483}{16} \left(X + \frac{2137}{2322} \right)^2 + \frac{131797}{5418} > 0. \end{aligned}$$

Likewise

$$\begin{aligned} B_3 + 252 &= 6 \left(\nu + \frac{5}{3}X - 2Y - \frac{1}{2} \right)^2 + 24 \left(Y - \frac{11}{12}X - \frac{15}{4} \right)^2 \\ &\quad + \frac{139}{6} \left(X - \frac{249}{139} \right)^2 + \frac{1851}{278} > 0. \end{aligned}$$

Further elementary completions give

$$B_1 \geq \frac{112}{3}\nu \geq 0, \quad B_4 \geq \frac{132}{7} > 0.$$

B.4. The conditional middle-coefficient certificate. Set

$$\Sigma = 4B_2B_4 + 252B_3.$$

An exact Gram decomposition is

$$(B.64) \quad \Sigma = 2128\nu \left(\nu + \frac{387}{133}X - \frac{261}{133}Y + \frac{77}{20} \right)^2 + \frac{1}{59850}m^T Q m,$$

where $m = (1, Y, Y^2, X, XY, X^2)^T$ and

$$Q = \begin{pmatrix} 8739057600 & -4231634400 & 478800000 & 6265576800 & -1795500000 & 47880000 \\ -4231634400 & 2519799114 & -385003080 & -2526627600 & 807975000 & 239400000 \\ 478800000 & -385003080 & 79782560 & 281630160 & -81252000 & -35910000 \\ 6265576800 & -2526627600 & 281630160 & 10931106942 & -2359670040 & 2303554680 \\ -1795500000 & 807975000 & -81252000 & -2359670040 & 685371360 & -243756000 \\ 47880000 & 239400000 & -35910000 & 2303554680 & -243756000 & 1122611040 \end{pmatrix}.$$

Exact rational LDL^T elimination has positive pivots

$$\begin{array}{l} 8739057600, \quad \frac{79557301866}{169}, \quad \frac{1112696849800960}{299087601}, \\ \frac{2624099206828303859949}{869294413907}, \quad \frac{2531308970961028469920080}{115380521779374043}, \\ \frac{1406694633431943161674826341030}{10547120712337618624667}. \end{array}$$

Thus $Q \succ 0$ and $m^T Q m > 0$ because the first coordinate of m is one. Equation (B.64) proves $\Sigma > 0$.

If $B_3 \geq 0$, then $B_2 + aB_3 + a^2B_4 > 0$ is immediate. If $B_3 < 0$, the identity

$$4B_2B_4 - B_3^2 = (4B_2B_4 + 252B_3) + (-B_3)(B_3 + 252)$$

shows that the discriminant of the quadratic $B_2 + aB_3 + a^2B_4$ is negative. Hence that quadratic is positive for every $a > 0$. This completes the proof that $N_1 > 0$.

Appendix C. Formal verification.

The theorem has been formalized in Lean 4.19.0 with Mathlib 4.19.0. The formal development covers the subgroup and character enumeration, the cycle expansions, the residual normalization identity, the rational Gram certificate, and the passage from correlation matrices to the full positive-semidefinite cone. The public theorem is `normalized_permanental_dominance_n4` in the module `PermanentalDominance.N4.NormalizedPermanentalDominance`. A detailed module map and an axiom audit are included with the source. In the module `A4GeometricHessian`, the machine-checked counterparts of the three core coefficient identities are as follows:

- `hessianDeterminant_expansion` for (5.45);
- `minimumNumerator_expansion` for (5.46);
- `collinearNumerator_scaled_identity` for (5.49).

The matrix denoted by Θ here is named `T` in that module. The rank-one identity and the exact interior minimum are theorems `geometricP_collinear_identity` and `minimumPoint_value`, located in `A4GeometricRankOne` and `A4GeometricMinimum`, respectively. These references locate the machine-checked counterparts; the identities and sign certificates used by the mathematical proof remain displayed above.

The formal source used here was extracted from the immutable merge commit [c5e4d8c61f08](#) and is mirrored in the companion repository.

Acknowledgments. During the preparation of this work, the author used OpenAI ChatGPT and Codex to assist with mathematical exploration, development of the Lean formalization, literature organization, manuscript drafting, language editing, and L^AT_EX preparation. The author reviewed and edited all generated content and takes full responsibility for the accuracy and integrity of the work.

Data and code availability. No external data were used in this study. The accompanying Lean formalization is available from the [companion repository](#), together with the manuscript source. The immutable formalization provenance is the commit identified in the preceding section.

Funding and competing interests. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors. The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

REFERENCES

- [1] W. BARRETT, H. T. HALL, AND R. LOEWY, *The cone of class function inequalities for the 4-by-4 positive semidefinite matrices*, Proceedings of the London Mathematical Society, 79 (1999), pp. 107–130, <https://doi.org/10.1112/S0024611599011909>.
- [2] G.-S. CHEON AND I. M. WANLESS, *An update on Minc’s survey of open problems involving permanents*, Linear Algebra and its Applications, 403 (2005), pp. 314–342, <https://doi.org/10.1016/j.laa.2005.02.030>.
- [3] S. GELLER AND E. KNILL, *Measuring multiparticle indistinguishability with the generalized bunching probability*, Physical Review A, 113 (2026), p. 042606, <https://doi.org/10.1103/1x5l-jd26>.
- [4] P. HEYFRON, *Immanant dominance orderings for hook partitions*, Linear and Multilinear Algebra, 24 (1988), pp. 65–78.
- [5] G. D. JAMES, *Immanants*, Linear and Multilinear Algebra, 32 (1992), pp. 197–210, <https://doi.org/10.1080/03081089208818163>.
- [6] N. JING, Y. LIU, AND J. ZHANG, *Immanant inequalities and weight spaces*, 2025, <https://doi.org/10.48550/arXiv.2508.20382>, <https://arxiv.org/abs/2508.20382>.
- [7] E. H. LIEB, *Proofs of some conjectures on permanents*, Journal of Mathematics and Mechanics, 16 (1966), pp. 127–134, <https://doi.org/10.1512/iumj.1967.16.16008>.
- [8] M. MARCUS, *The permanent analogue of the Hadamard determinant theorem*, Bulletin of the American Mathematical Society, 69 (1963), pp. 494–496, <https://doi.org/10.1090/S0002-9904-1963-10975-1>.
- [9] R. MERRIS, *The permanental dominance conjecture*, in Current Trends in Matrix Theory, R. Grone and F. Uhlig, eds., North-Holland, New York, 1987, pp. 213–223.
- [10] R. MERRIS AND W. WATKINS, *Inequalities and identities for generalized matrix functions*, Linear Algebra and its Applications, 64 (1985), pp. 223–242, [https://doi.org/10.1016/0024-3795\(85\)90279-4](https://doi.org/10.1016/0024-3795(85)90279-4).
- [11] T. H. PATE, *Partitions, irreducible characters, and inequalities for generalized matrix functions*, Transactions of the American Mathematical Society, 325 (1991), pp. 875–894, <https://doi.org/10.1090/S0002-9947-1991-0998356-3>.
- [12] T. H. PATE, *Row appending maps, Ψ -functions, and immanant inequalities for hermitian positive semi-definite matrices*, Proceedings of the London Mathematical Society, 76 (1998), pp. 307–358, <https://doi.org/10.1112/S0024611598000100>.
- [13] T. H. PATE, *Tensor inequalities, ξ -functions and inequalities involving immanants*, Linear Algebra and its Applications, 295 (1999), pp. 31–59, [https://doi.org/10.1016/S0024-3795\(99\)00035-X](https://doi.org/10.1016/S0024-3795(99)00035-X).
- [14] K. RODTES, *Some remarks on permanental dominance conjecture*, Advances in Applied Mathematics, 160 (2024), p. 102758, <https://doi.org/10.1016/j.aam.2024.102758>.
- [15] I. SCHUR, *Über endliche gruppen und hermitesche formen*, Mathematische Zeitschrift, 1 (1918), pp. 184–207, <https://doi.org/10.1007/BF01465095>.
- [16] G. W. SOULES, *An approach to the permanental-dominance conjecture*, Linear Algebra and its Applications, 201 (1994), pp. 211–229, [https://doi.org/10.1016/0024-3795\(94\)90117-1](https://doi.org/10.1016/0024-3795(94)90117-1).

- [17] R. TABATA, *The values of the generalized matrix functions of 3×3 matrices*, Hiroshima Mathematical Journal, 45 (2015), pp. 1–8, <https://doi.org/10.32917/hmj/1428365051>.
- [18] I. M. WANLESS, *Lieb's permanental dominance conjecture*, in The Physics and Mathematics of Elliott Lieb: The 90th Anniversary, vol. 2, EMS Press, Berlin, 2022, pp. 501–516, <https://doi.org/10.4171/90-2/48>.
- [19] F. ZHANG, *An update on a few permanent conjectures*, Special Matrices, 4 (2016), pp. 305–316, <https://doi.org/10.1515/spma-2016-0030>.